

EAGNA

An Integrated Virtual Learning Environment

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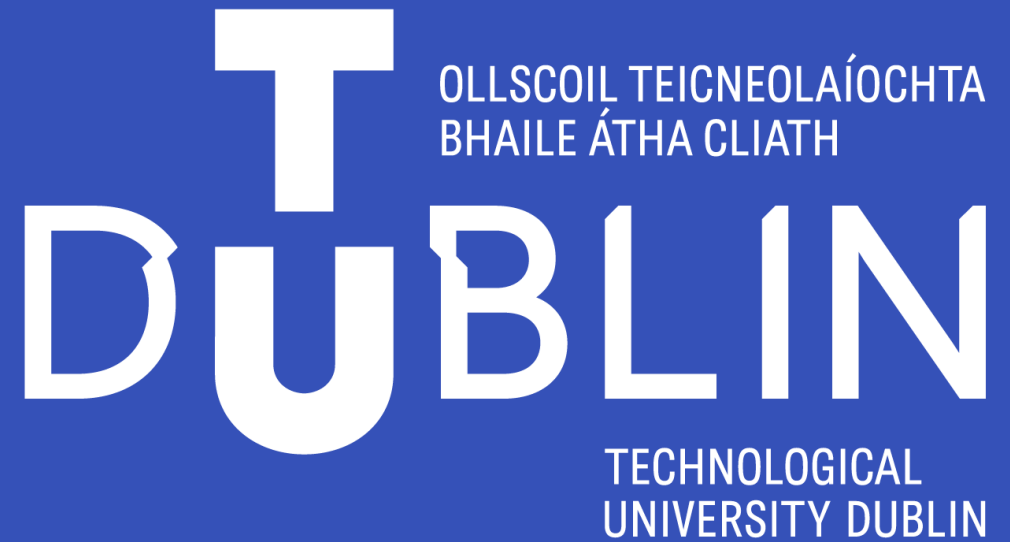
Supervisor – Eoin Rogers

Overview

Eagna is a condensed **Virtual Learning Environment** designed for the Technological University of Dublin.

Features:

- **Login** for Student & Lecturers.
- **Registration** for Students.
- **Modules & Assignments.**
- **Admin & External Tools.**



Why?

This project was designed to deal with the **registration & Brightspace enrolment** problem.

Advantages:

- **Built to be hosted on-site or on-cloud.**
- **Change features on-demand.**
- **Control over security & management of sensitive data.**

SN	Student News	Student Registration Update: System Unavailable Today from 19:...	Mon 10/11
		Planned maintenance for online registration this Mon, 10 Nov 19...	Inbox
> TU	TU Dublin Update	Update on Registration - 17 October 2025	17/10/2025
		Dear Students, We understand that many of you are still experienc...	Inbox
> TU	TU Dublin Update	Update on Registration	26/09/2025
		Dear Student, Thank you for your patience as we introduce our ne...	Inbox
> T	TU Dublin, Computer Science	Registration for upcoming academic year	12/09/2025
		Dear Student, I hope you enjoyed your summer break, are refresh...	Inbox
> TU	TU Dublin Update	A message from President Deirdre Lillis: Update on Registration..	10/10/2025
		Update on Registration Issues and What We Are Doing to Fix The...	Inbox
> TU	TU Dublin Update	Important Update on Registration	12/09/2025
		Dear Student, We're excited to welcome you to TU Dublin and can...	Inbox
> TU	TU Dublin Update	Update on Registration 2025/26	25/08/2025
		Important Update about Registration Dear Student, As per the em...	Inbox

Research

Extensive research was performed to understand VLEs.

Including:

- Existing VLEs (Brightspace etc.)
- Technologies (Django etc.)
- Requirements (Accessibility etc.)
- Existing Final Year Projects

My Home Progress ▾ Discover Calendar Help ▾

My Modules

All 2020-21 2021-22 2022-23 2023-24 2024-25 2025-2



TU856, TU857, TU858
**ADVANCED
DATABASES**

Advanced Databases
CMPU4003: Sept-Dec
Academic Term
2025/26



Rich Web Application
Tech CMPU4043:
Sept-Dec Academic
Term 2025/26

For
Sep
Term



Requirements

Feedback collection & requirements gathering performed to **understand user needs.**

Including:

- Questionnaires
- Brightspace Feature Review
- Use Case Scenarios
- Requirements Specification

Accessibility

6. Do you identify with any of the following disability categories? *

- ☐ No
- ☐ Yes - Prefer Not To Specify
- ☐ Visual Impairment
- ☐ Hearing Impairment
- ☐ Mobility Impairment
- ☐ Speech Impairment
- ☐ Learning Disability
- ☐ Neurodivergent
- ☐ Long Term Illness
- ☐ Mental Health Condition
- ☐ Other

Design

Feature Driven Development was chosen as the methodology.

The design of Eagna has been outlined based on the following:

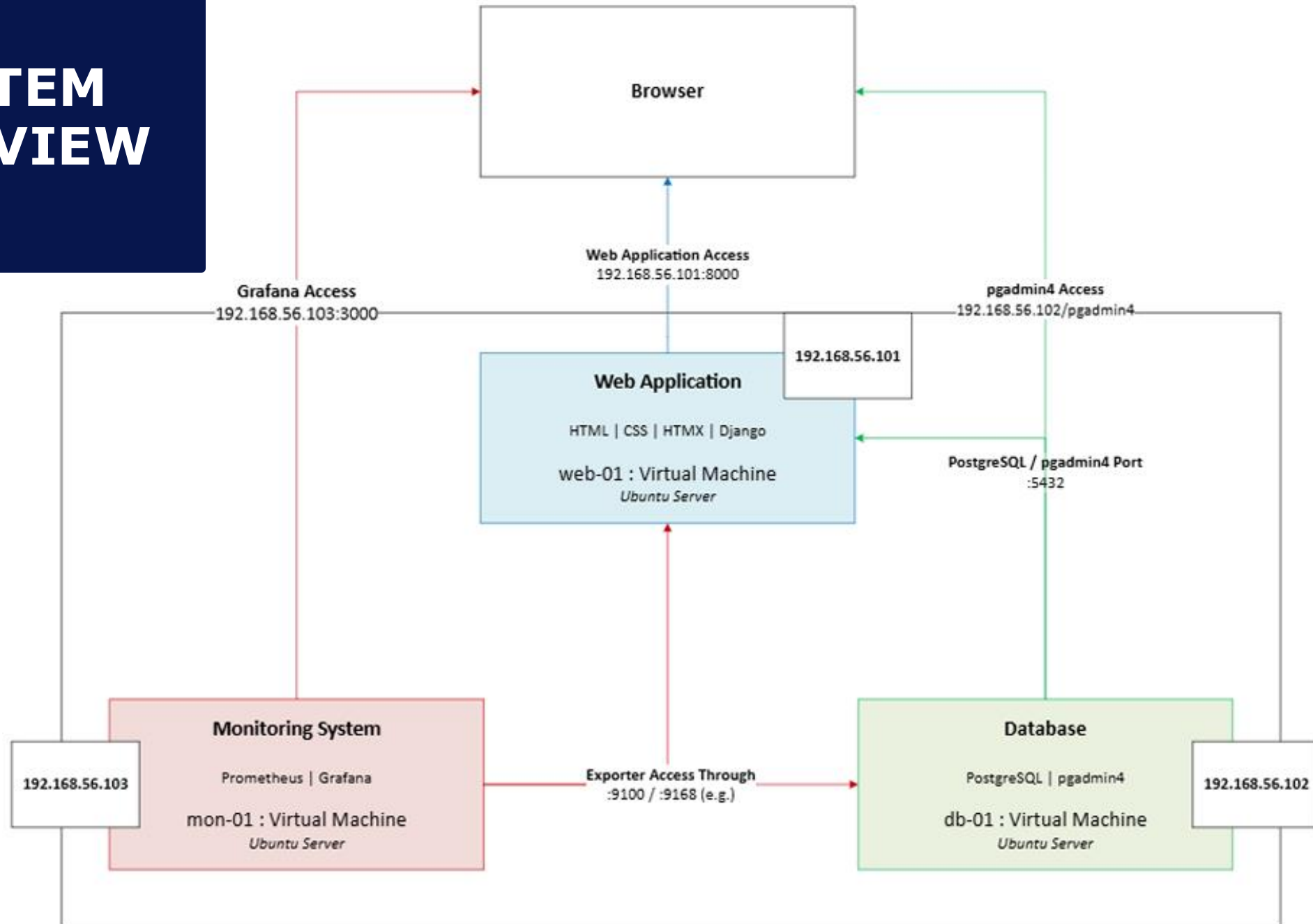
- **Dependencies**
- **Complexity**
- **Risk**

Wireframes & Diagrams have been used to visualize the design.

The wireframe illustrates the 'STUDENT ASSIGNMENTS' page. It features a dark blue header with the 'EAGNA' logo and navigation links: 'DASHBOARD | TOOLS | INBOX | WEBSITE'. On the right of the header are a bell icon, 'USER', and 'LOGOUT'. The main content area is divided into several sections: 1. Assignment details: 'ASSIGNMENT TITLE - MODULE' and 'DUE DATE / TIME - MAX MARK'. 2. Description: A large text input field labeled 'DESCRIPTION'. 3. Materials: A text input field labeled 'ASSIGNMENT MATERIALS'. 4. Submission section: Titled 'YOUR SUBMISSION', it contains a box for 'STATUS' (with sub-labels 'SUBMISSION TIMESTAMP' and 'SUBMISSION FILES'), a box for 'GRADE' (with sub-labels 'MARK' and 'FEEDBACK'), and a 'SUBMIT WORK' section. The 'SUBMIT WORK' section includes a 'CHOOSE FILE' button and a prominent 'SUBMIT' button. 5. Footer: A dark blue bar with the 'TUD LOGO' and navigation links 'DASHBOARD | TOOLS | INBOX | WEBSITE'.

Figure 111 - Wireframes - Student Assignments – Blown Up

SYSTEM OVERVIEW

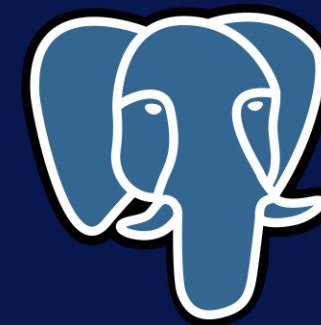


Implementation

These **technologies** were chosen to build the system:

- **Frontend:** HTML, CSS, HTMX (JS)
- **Backend:** Django
- **Database:** PostgreSQL
- **Virtualization:** VirtualBox
(*Ubuntu Server 24.04 LTS*)
- **Monitoring:** Prometheus & Grafana

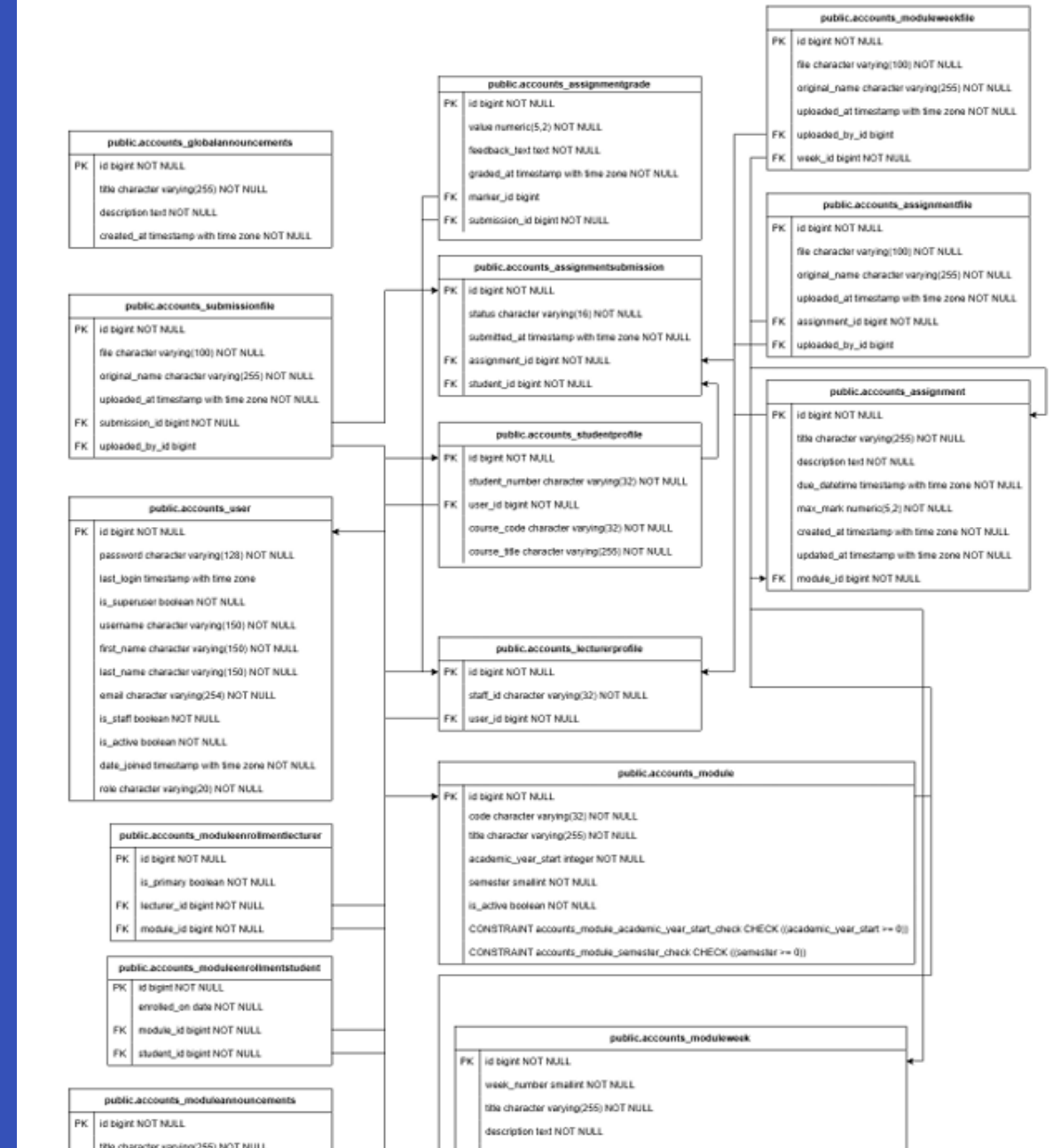
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Complexity

With **three** interconnected virtual machines running a **variety of technologies** – the complexity in this project is **significant**.

- **Rich Role-Based Domain Logic & Access Control**
- **Distributed & Database Backed Architecture**
- **Concurrent User Support**
- **Dynamic User Interface**



Testing & Evaluation

Numerous different **testing** and **evaluation methods** will be used.

Testing:

- Black / White Box Testing
- Unit Testing

Evaluation:

- Nielsen's Heuristics
- User Feedback
- TPC / CIS Benchmarks
- Four Golden Signals

Lecturer User

Test ID	Functionality	Steps	Expected Results
W-L-01	Login	❖ Start At Login Page. ❖ Enter Login Details. ❖ Click Login.	User gains access to Dashboard.
W-L-02	Module Access	❖ Start At Dashboard ❖ Click on Module.	User gains access to Module.
W-L-03	Assignment Access	❖ Start At Dashboard ❖ Click on Assignment.	User gains access to Assignment.
W-L-04	Academic Calendar Editing	❖ Start At Dashboard ❖ Click On Module ❖ Go to any week in the Academic Calendar. ❖ Update Description. ❖ Attach File.	Student users should be able to gain access and view updated description / attached files.
W-L-05	Create Assignment	❖ Start At Dashboard ❖ Click Module ❖ Click Create Assignment ❖ Fill Out Details ❖ Submit	A new assignment should be visible within the module page.
W-L-06	Add Grade	❖ Start At Dashboard ❖ Click on assignment where student has submitted. ❖ Click Grade Button. ❖ Fill out details.	Grade should now be available on submission.
W-L-07	Create Module Announcement	❖ Start At Dashboard. ❖ Go To A Module. ❖ Create Module Announcement.	A new announcement should be visible within the module page.

Assignment 2 – CMPU4043 Rich Web Application Technology

Due: 2025-12-12 16:00 · Max mark: 100.00

Description

Four Phased Assignment

Assignment Materials

ca2_start_files.zip (uploaded 2025-11-20 11:09)

Your Submission

Status: Submitted

Submitted: 2025-11-20 11:15

Submitted files

current.pdf (uploaded 2025-11-20 11:15)

Grade

Mark: 70.00 / 100.00

Feedback: 100

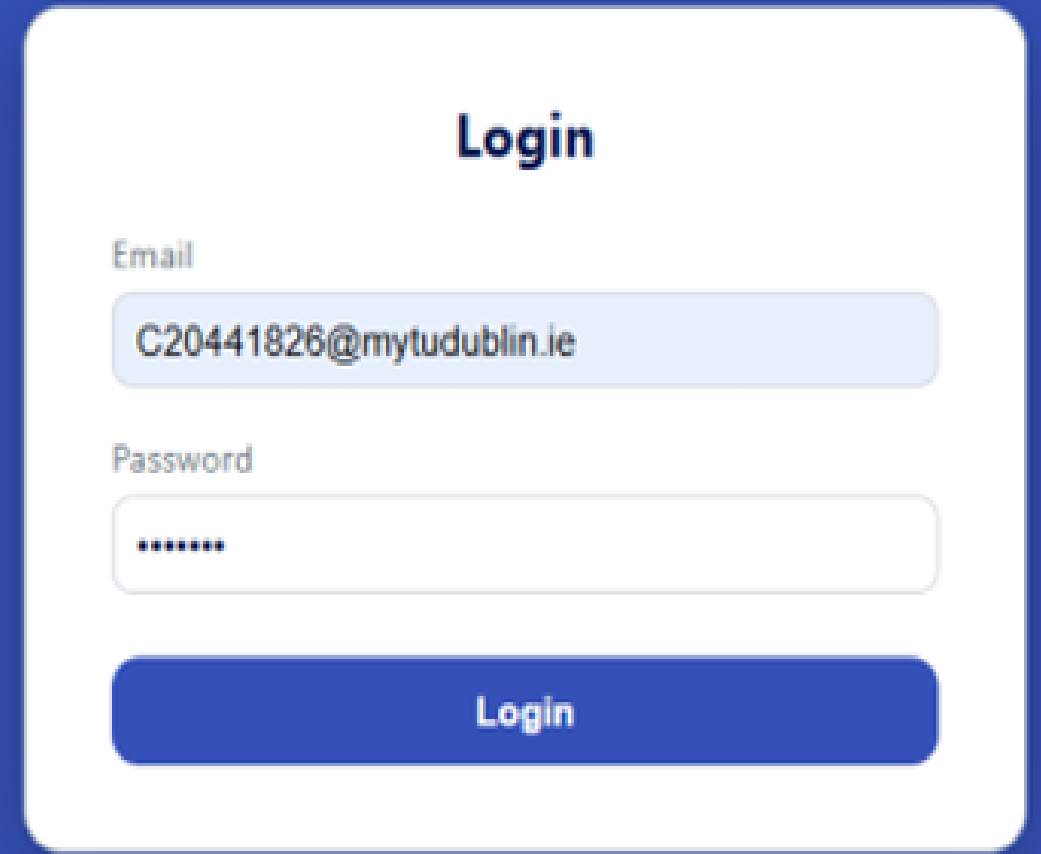
Submit work

THE PROTOTYPE

Interim Features

Student & Lecturer

- Login System (& Logout)
- Module Functionality
- Assignment Functionality
- Navigation Menus & Links

A mockup of a login form. It has a white background with rounded corners, set against a dark blue background. At the top, the word "Login" is written in a bold, dark blue font. Below this, there are two input fields. The first is labeled "Email" in a small, grey font, and contains the text "C20441826@mytudublin.ie". The second is labeled "Password" in a small, grey font, and contains a series of dots representing a masked password. Below these fields is a large, rounded blue button with the word "Login" in white text.

Login

Email

C20441826@mytudublin.ie

Password

Login

TIME FOR THE DEMO

THANK YOU.

Any Questions?